Chapter 19

Campbell Biology, AP Edition, 12e

Micrograph showing human immune cell releasing scores of human immunodeficiency viruses.

Chapter 19

Viruses

Figure 19.1

A human immune cell (blue) infected by human immunodeficiency virus (HIV) is releasing more HIV viruses (gold dots), which will go on to infect other cells. Left untreated, HIV destroys vital immune system cells, causing AIDS. This disease has killed about 35 million people worldwide.

A diagram of virus replication. How does a virus makes more virus?

A virus consists only of nucleic acid, proteins, and sometimes a membranous envelope. After infecting a host cell, it uses the host cell's molecules to make new viruses, as shown in this simplified diagram. A generic virus containing a viral genome, D N A or R N A, enters a host cells and releases the genome. The viral genome is replicated by host enzymes. Copies of the genome are transcribed by host enzymes. Viral m R N A is translated by host ribosomes. Viral proteins are self assembled with the viral genome. New viruses are released from the cell. With some viruses, the host cell bursts when the new viruses are released. With some viruses, the host cell remains alive after the new viruses are released.

Key Concepts

- **19.1****A virus consists of a nucleic acid surrounded by a protein coat**
- **19.2****Viruses replicate only in host cells**
- **19.3****Viruses and prions are formidable pathogens in animals and plants**

Advanced placement edition

Viral replication is unique in that it is dependent on the cellular machinery of its host **(BIG IDEA 3)**. Viruses can also introduce variation into the genetic material of a host, which can lead to rapid evolutionary

change **(BIG IDEA 1)**. Disruption to ecosystems can expose plants and animals to different viruses (**BIG IDEA 4)**.

Study Tip

Make flowcharts: Make a simple flowchart for each viral replicative cycle in this chapter. Below is the beginning of an example for the lysogenic cycle. Note the similarities and differences between the replicative cycles.

The lysogenic cycle. Virus attaches to bacterial cell and injects genome. Viral genome is integrated into the bacterial chromosome.

Get Ready for This Chapter

In this chapter, you will learn about how viruses were discovered, their structure and means of replication, and their impact in terms of disease. Prions as infectious proteins are also described. Before beginning this chapter, you should be able to do the following:

- Distinguish between the structures of a bacterial cell and a eukaryotic cell (see Concept 6.2, Figure 6.5, and Figure 6.8).
- Explain how two proteins bind to each other (see Concept 5.4).
- Describe the process of gene expression (see Concept 17. 2, Concept 17.4, and Figure 17.4).

The following questions provide a quick check of your basic knowledge in these areas.

1. Which of the following clues would tell you if a cell is prokaryotic or eukaryotic?

```
the presence or absence of ribosomes
```

That's incorrect. Try again.

Incorrect. Review the following material: All cells are bounded by a selective barrier called the plasma membrane, and all cells have ribosomes, tiny complexes that make proteins according to the instructions from DNA. Eukaryotic cells, however, are generally much larger than prokaryotic cells and have internal membranes, which are not present in most prokaryotic cells. In a eukaryotic cell, most of the DNA is in an organelle called the nucleus, whereas in prokaryotic cells, the DNA is concentrated in a region that is not membrane-enclosed, called the nucleoid.

```
whether or not the cell carries out cellular metabolism
```

That's incorrect. Try again.

Incorrect. Review the following material: All cells are bounded by a selective barrier called the plasma membrane, and all cells have ribosomes, tiny complexes that make proteins according to the instructions from DNA. Eukaryotic cells, however, are generally much larger than prokaryotic cells and have internal membranes, which are not present in most prokaryotic cells. In a eukaryotic cell, most of the DNA is in an organelle called the nucleus, whereas in prokaryotic cells, the DNA is concentrated in a region that is not membrane-enclosed, called the nucleoid.

```
whether or not the cell is partitioned by internal membranes
```

That's correct.

Correct! If you had trouble with this question, review the following material: All cells are bounded by a selective barrier called the plasma membrane, and all cells have ribosomes, tiny complexes that make proteins according to the instructions from DNA. Eukaryotic cells, however, are generally much larger than prokaryotic cells and have internal membranes, which are not present in most prokaryotic cells. In a eukaryotic cell, most of the DNA is in an organelle called the nucleus, whereas in prokaryotic cells, the DNA is concentrated in a region that is not membrane-enclosed, called the nucleoid.

```
the presence or absence of a rigid cell wall
```

That's incorrect. Try again.

Incorrect. Review the following material: All cells are bounded by a selective barrier called the plasma membrane, and all cells have ribosomes, tiny complexes that make proteins according to the instructions from DNA. Eukaryotic cells, however, are generally much larger than prokaryotic cells and have internal membranes, which are not present in most prokaryotic cells. In a eukaryotic cell, most of the DNA is in an organelle called the nucleus, whereas in prokaryotic cells, the DNA is concentrated in a region that is not membrane-enclosed, called the nucleoid.

2. What determines whether two proteins will bind to each other?

```
        They will bind to each other if they are located in the
same region of the cell.
```

That's incorrect. Try again.

Incorrect. Review the following material: A protein's specific structure determines how it works, and in almost every case, the function of a protein depends on its ability to recognize and bind to some other molecule. This fit is very specific, something like the fit of two hands in a handshake, and depends on both the shape and the overall chemical properties of the regions where the proteins might bind to each other.

```
        They will bind to each other if one protein has an
overall positive charge and one protein has an overall negative
charge.
```

That's incorrect. Try again.

Incorrect. Review the following material: A protein's specific structure determines how it works, and in almost every case, the function of a protein depends on its ability to recognize and bind to some other molecule. This fit is very specific, something like the fit of two hands in a handshake, and depends on both the shape and the overall chemical properties of the regions where the proteins might bind to each other.

```
        They will bind to each other if they have complementary
shapes and chemical properties.
```

That's correct.

Correct! If you had trouble with this question, review the following material: A protein's specific structure determines how it works, and in almost every case, the function of a protein depends on its ability to recognize and bind to some other molecule. This fit is very specific, something like the fit of two hands in a handshake, and depends on both the shape and the overall chemical properties of the regions where the proteins might bind to each other.

```
        They will bind to each other if they are identical or
similar proteins.
```

That's incorrect. Try again.

Incorrect. Review the following material: A protein's specific structure determines how it works, and in almost every case, the function of a protein depends on its ability to recognize and bind to some other molecule. This fit is very specific, something like the fit of two hands in a handshake, and depends on both the shape and the overall chemical properties of the regions where the proteins might bind to each other.

3. Which of the following events occurs during the process of transcription?

```
An RNA is synthesized.
```

That's correct.

Correct! If you had trouble with this question, review the following material: Transcription is the synthesis of RNA using the information in DNA. Translation is the synthesis of a polypeptide using the information in mRNA. The sites of translation are ribosomes, molecular complexes that facilitate the orderly linking of amino acids into polypeptide chains.

```
An mRNA attaches to a ribosome.
```

That's incorrect. Try again.

Incorrect. Review the following material: Transcription is the synthesis of RNA using the information in DNA. Translation is the synthesis of a polypeptide using the information in mRNA. The sites of translation are ribosomes, molecular complexes that facilitate the orderly linking of amino acids into polypeptide chains.

```
DNA is replicated.
```

That's incorrect. Try again.

Incorrect. Review the following material: Transcription is the synthesis of RNA using the information in DNA. Translation is the synthesis of a polypeptide using the information in mRNA. The sites of translation are ribosomes, molecular complexes that facilitate the orderly linking of amino acids into polypeptide chains.

```
A polypeptide is synthesized.
```

That's incorrect. Try again.

Incorrect. Review the following material: Transcription is the synthesis of RNA using the information in DNA. Translation is the synthesis of a polypeptide using the information in mRNA. The sites of translation

are ribosomes, molecular complexes that facilitate the orderly linking of amino acids into polypeptide chains.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 19.1 Describe a virus.
- 19.2 Explain how viruses replicate and evolve.
- 19.3 Use examples to illustrate the effects of animal and plant viruses, as well as prions, on their hosts.

Concept 19.1: A virus consists of a nucleic acid surrounded by a protein coat

For the past few years, world news has focused on the virus called **SARS-CoV-2**, short for Severe Acute Respiratory Syndrome-CoronaVirus-2, which causes **COVID-19**. This virus first infects cells in the airway, where it produces more viruses that can be transmitted to other individuals via exhaled breath. Worldwide, COVID-19 has killed more than 6.5 million individuals by late 2022. (The actual number is likely significantly higher as many people who died from COVID-19-like symptoms were not tested for the virus.)

What distinguishes SARS-CoV-2 and other viruses from cells? Compared with eukaryotic and even prokaryotic cells, most viruses are much smaller and simpler in structure. Lacking the structures and metabolic machinery found in a cell, a **virus** is an infectious particle consisting of little more than genes packaged in a protein coat.

Are viruses living or nonliving? Early on, they were considered biological chemicals; the Latin root for *virus* means "poison." Viruses can cause a wide variety of diseases, so researchers in the late 1800s saw a parallel with bacteria and proposed that viruses were the simplest of living forms. However, viruses cannot reproduce or carry out metabolic activities outside of a host cell. Most biologists would probably agree that viruses are not alive, but instead exist in a shady area between life-forms and chemicals. The simple phrase used by two researchers describes them aptly: Viruses lead "a kind of borrowed life."

The Discovery of Viruses: *Scientific Inquiry*

Scientists detected viruses indirectly long before they were able to see them. The story of how viruses were discovered begins in 1883. A German scientist named Adolf Mayer was studying tobacco mosaic disease,

which stunts the growth of tobacco plants and gives their leaves a mottled, or mosaic, coloration. Mayer discovered that he could transmit the disease from plant to plant by rubbing sap extracted from diseased leaves onto healthy plants. After an unsuccessful search for an infectious microorganism in the sap, he suggested that the disease was caused by unusually small bacteria that were invisible under a microscope. This hypothesis was tested a decade later by Dmitri Ivanowsky, a Russian biologist who passed sap from infected tobacco leaves through a filter designed to remove bacteria. After filtration, the sap still produced mosaic disease.

But Ivanowsky reasoned that perhaps the bacteria were small enough to pass through the filter or made a toxin that could do so. The second possibility was ruled out when the Dutch botanist Martinus Beijerinck carried out a classic series of experiments that showed that the infectious agent in the filtered sap could replicate **(Figure 19.2)**.

Figure 19.2

Inquiry: What causes tobacco mosaic disease?

Experiment

In the late 1800s, Martinus Beijerinck, of the Technical School in Delft, the Netherlands, investigated the properties of the agent that causes tobacco mosaic disease (then called spot disease).

A diagram of the experiment.

The experiment is as follows.

1. Extracted sap from tobacco plant with tobacco mosaic disease.
2. Passed sap through a porcelain filter known to trap bacteria.
3. Rubbed filtered sap on healthy tobacco plants.
4. Healthy plants became infected.

Results

When the filtered sap was rubbed on healthy plants, they became infected. Their sap, extracted and filtered, could then act as a source of infection for another group of plants. Each successive group of plants developed the disease to the same extent as earlier groups.

Conclusion

The infectious agent was apparently not a bacterium because it could pass through a bacterium-trapping filter. The pathogen must have been replicating in the plants because its ability to cause disease was undiluted after several transfers from plant to plant.

Data from M. J. Beijerinck, Concerning a *contagium vivum fluidum* as cause of the spot disease of tobacco leaves, *Verhandelingen der Koninkyke akademie Wettenschappen te Amsterdam* 65:3–21 (1898). Translation published in English as Phytopathological Classics Number 7 (1942), American Phytopathological Society Press, St. Paul, MN.

WHAT IF? *If Beijerinck had observed that the infection of each group was weaker than that of the previous group and that ultimately the sap could no longer cause disease, what might he have concluded?*

*For suggested answer, see*Appendix A.**

In fact, the pathogen replicated only within the host it infected. In further experiments, Beijerinck showed that unlike bacteria used in the lab at that time, the mysterious agent of mosaic disease could not be cultivated on nutrient media in test tubes or petri dishes. Beijerinck imagined a replicating particle much smaller and simpler than a bacterium, and he is generally credited with being the first scientist to voice the concept of a virus. His suspicions were confirmed in 1935 when the American scientist Wendell Stanley crystallized the infectious particle, now known as tobacco mosaic virus (TMV). Subsequently, TMV and many other viruses were actually seen with the help of the electron microscope.

Structure of Viruses

The tiniest viruses are only 20 nm in diameter—smaller than a ribosome. Millions could easily fit on a pinhead. Even the largest known virus, which has a diameter of 1,500 nm (1.5 μm), is barely visible under the light microscope. Stanley's discovery that some viruses could be crystallized was exciting and puzzling news. Not even the simplest of cells can aggregate into regular crystals. But if viruses are not cells, then what are they? Examining the structure of a virus more closely reveals that it is an infectious particle consisting of one or more molecules of a nucleic acid enclosed in a protein coat and, for some viruses, surrounded by a membranous envelope.

The simple structure of viruses make them a useful biological system: To a large extent, molecular biology was born in the laboratories of biologists studying viruses that infect bacteria. Experiments using these viruses provided evidence that genes are made of nucleic acids, and they were critical in working out the molecular mechanisms of the fundamental processes of DNA replication, transcription, and translation.

Viral Genomes

We usually think of genes as being made of double-stranded DNA, but many viruses defy this convention. Their genomes may consist of double-stranded DNA, single-stranded DNA, double-stranded RNA, or single-stranded RNA, depending on the type of virus. A virus is called a DNA virus or an RNA virus based on the kind of nucleic acid that makes up its genome. In either case, the genome is usually organized as a single linear or circular molecule of nucleic acid, although the genomes of some viruses consist of multiple molecules of nucleic acid. The smallest viruses known have only three genes in their genome, while the largest have several hundred to 2,000. For comparison, bacterial genomes contain about 200 to a few thousand genes.

Capsids and Envelopes

The protein shell enclosing the viral genome is called a **capsid**. Depending on the type of virus, the capsid may be rod-shaped, polyhedral, or more complex in shape. Capsids are built from a large number of protein subunits called *capsomeres*, but the number of different *kinds* of proteins in a capsid is usually small. Tobacco mosaic virus has a rigid, rod-shaped capsid made from over 1,000 molecules of a single type of protein arranged in a helix; rod-shaped viruses are commonly called *helical viruses* for this reason **(Figure 19.3a)**. Adenoviruses, which infect the respiratory tracts of animals, have 252 identical protein molecules arranged in a polyhedral capsid with 20 triangular facets—an icosahedron; thus, these and other similarly shaped viruses are referred to as *icosahedral viruses***(Figure 19.3b)**.

Figure 19.3

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Viral structure.
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Diagrams and micrographs of four viral structures.

Viruses are made up of nucleic acid (DNA or RNA) enclosed in a protein coat (the capsid) and sometimes further wrapped in a membranous envelope. The individual protein subunits making up the capsid are called capsomeres. Although diverse in size and shape, viruses have many common structural features. (All micrographs are colorized TEMs.)

The structures are as follows.

Part a. Tobacco mosaic virus has a helical capsid surrounding a single helical R N A molecule, with the overall shape of a rigid rod. Scale is 18 by 250 nanometers. Capsomeres are egg shaped.

Part b. Adenoviruses have an icosahedral capsid that encloses the D N A with a protein spike at each vertex (corner). The capsid, made up of capsomeres, contains double-stranded D N A. Scale: 70 to 90 nanometers (diameter).

Part c. Influenza viruses have an outer envelope, studded with glycoprotein spikes, that contains eight double-helical R N A-protein complexes, each associated with a viral polymerase. Scale: 80 to 200 nanometers (diameter).

Part d. Bacteriophage T4, like other "T-even" phages, has a complex capsid consisting of an icosahedral head and a tail apparatus, consisting of the tail sheath (cylinder) and tail fibers. The double-stranded D N A is housed in the head. Scale: 80 by 225 nanometers.

Some viruses have accessory structures that help them infect their hosts. For instance, a membranous envelope surrounds the capsids of influenza viruses, coronaviruses, and many other viruses found in animals **(Figure 19.3candd)**. These **viral envelopes**, which are derived from the membranes of the host cell, contain host cell phospholipids and membrane proteins. They also contain proteins and glycoproteins of viral origin. (Glycoproteins are proteins with carbohydrates covalently attached.) Some viruses carry a few viral enzymes, such as viral polymerase, within their capsids.

Many of the most complex capsids are found among the viruses that infect bacteria, called **bacteriophages**, or simply ****phages**. The first phages studied included seven that infect *Escherichia coli* (*E. coli*). These seven phages were named type 1 (T1), type 2 (T2), and so forth, in the order of their discovery. (The T2 phage was used in the experiment that established DNA as the genetic material; see Figure 16.4.) The three “T-even” phages (T2, T4, and T6) turned out to be very similar in structure. Their capsids have elongated icosahedral heads enclosing their DNA. Attached to the head is a protein tail piece with fibers by which the phages attach to a bacterial cell **(Figure 19.3d)**. In the next section, we’ll examine how these few viral parts function together with cellular components to produce large numbers of viral progeny.

Concept Check 19.1

- **VISUAL SKILLS** Describe one similarity and two differences between the structures of tobacco mosaic virus (TMV) and influenza virus (see Figure 19.3).
- **MAKE CONNECTIONS** Bacteriophages were used to provide evidence that DNA carries genetic information (see Figure 16.4). Briefly describe the experiment carried out by Hershey and Chase, including in your description why the researchers chose to use phages.

*For suggested answers, see*Appendix A.**

Concept 19.2: Viruses replicate only in host cells

Viruses lack metabolic enzymes and equipment for making proteins, such as ribosomes. They are obligate intracellular parasites; in other words, they can replicate only within a host cell. It is fair to say that viruses

in isolation are merely packaged sets of genes in transit from one host cell to another.

Each particular virus can infect cells of only a limited number of host species, called the **host range** of the virus. This host specificity results from the evolution of recognition systems by the virus. Viruses usually identify host cells by a “handshake” fit between viral surface proteins and specific receptor molecules on the outside of cells. Such receptor molecules are most often proteins that carry out necessary functions for the host cell and have been adopted by viruses as portals of entry. Some viruses have broad host ranges. For example, West Nile virus and equine encephalitis virus are distinctly different viruses that can each infect mosquitoes, birds, horses, and humans. Other viruses have host ranges so narrow that they infect only a single species. Measles virus, for instance, can infect only humans. Furthermore, viral infection of multicellular eukaryotes is usually limited to particular tissues. Human cold viruses infect only the cells lining the upper respiratory tract, and the HIV seen in Figure 19.1 binds to receptors present only on certain types of immune cells.

General Features of Viral Replicative Cycles

A viral infection begins when a virus binds to a host cell and the viral genome makes its way inside **(Figure 19.4)**. The mechanism of genome entry depends on the type of virus and the type of host cell. For example, T-even phages use their elaborate tail apparatus to inject DNA into a bacterium (see Figure 19.3d). Other viruses are taken up by endocytosis or, in the case of enveloped viruses, by fusion of the viral envelope with the host’s plasma membrane. Once the viral genome is inside, the proteins it encodes can commandeer the host, reprogramming the cell to copy the viral genome and manufacture viral proteins. The host provides the nucleotides for making viral nucleic acids, as well as enzymes, ribosomes, tRNAs, amino acids, ATP, and other components needed for making the viral proteins. Many DNA viruses use the DNA polymerases of the host cell to synthesize new genomes along the templates provided by the viral DNA. In contrast, to replicate their genomes, RNA viruses use virally encoded RNA polymerases that can use RNA as a template. (Uninfected cells generally make no enzymes for carrying out this process.)

Figure 19.4

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A simplified viral replicative cycle.
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Simplified viral replicative cycle; the parasite is a D N A virus with a capsid consisting of a single type of protein which enters the cell. New virus particles exit the cell.

A virus is an intracellular parasite that uses the equipment and small molecules of its host cell to replicate. In this simplest of viral cycles, the parasite is a DNA virus with a capsid consisting of a single type of protein.

Simplified viral replicative cycle; the parasite is a D N A virus with a capsid consisting of a single type of protein which enters the cell.

1. The virus enters the cell and is uncoated, releasing viral D N A and capsid proteins.
2. Host enzymes replicate the viral genome inside the cell.
3. Meanwhile, host enzymes inside the cell transcribe the viral genome into viral m R N A, which host ribosomes us to make more capsid proteins.
4. Viral genomes and capsid proteins self-assemble into new virus particles, which exit the cell.

MAKE CONNECTIONS *Label each of the straight gray arrows to identify the process that is occurring. Review**Figure 17.25.*

*For suggested answer, see*Appendix A.**

Animation

Simplified Viral Replicative Cycle

image

image

After the viral nucleic acid molecules and capsomeres are produced, they spontaneously self-assemble into new viruses. In fact, researchers can separate the RNA and capsomeres of TMV and then reassemble complete viruses simply by mixing the components together under the right conditions. The simplest type of viral replicative cycle ends with the exit of hundreds or thousands of viruses from the infected host cell, a process that often damages or destroys the cell. Such cellular damage and death, as well as the body's responses to this destruction, cause many of the symptoms associated with viral infections. The viral progeny that exit a cell have the potential to infect additional cells, spreading the viral infection.

There are many variations on the simplified viral replicative cycle we have just described. We will now take a look at some of these variations in bacterial viruses (phages) and animal viruses; later in the chapter, we will consider plant viruses.

Replicative Cycles of Phages

Phages are the best understood of all viruses, although some of them are also among the most complex. Research on phages led to the discovery that some double-stranded DNA viruses can replicate by two alternative mechanisms: the lytic cycle and the lysogenic cycle.

The Lytic Cycle

A phage replicative cycle that culminates in death of the host cell is known as a **lytic cycle**. The term *lytic* refers to the last stage of infection, during which the bacterium lyses (breaks open) and releases the phages that were produced within the cell. Each of these phages can then infect a healthy cell, and a few successive lytic cycles can destroy an entire bacterial population in just a few hours. A phage that replicates only by a lytic cycle is a **virulent phage**. **Figure 19.5** illustrates the major steps in the lytic cycle of T4, a typical virulent phage.

Figure 19.5

The lytic cycle of phage T4, a virulent phage.

The lytic cycle of phage T4: 1) Attachment. 2) Entry of phage D N A and degradation of host D N A. 3) Synthesis of viral genomes and proteins. 4) Self-assembly. 5) Release.

The lytic cycle of phage T4:

1. Attachment. The T4 phage uses its tail fibers to bind to specific surface proteins on an E. coli cell that act as receptors.
2. Entry of phase D N A and degradation of host D N A. The sheath of the tail contracts, injecting the phage D N A into the cell and leaving an empty capsid outside. The cell's D N A is hydrolyzed.
3. Synthesis of viral genomes and proteins. The phage D N A directs production of phage proteins and copies of the phage genome by host and viral enzymes, using components within the cell.
4. Self-assembly. Three separate sets of protein self-assemble to form phage heads, tails, and tail fibers. The phage genome is packaged inside the capsid as the head forms.
5. Release. The phage directs production of an enzyme that damages the bacterial wall, allowing fluid to enter. The cell swells and finally bursts, releasing 100 to 200 phage particles.

Phage T4 has almost 300 genes, which are transcribed and translated using the host cell's machinery. One of the first phage genes translated after the viral DNA enters the host cell codes for an enzyme that degrades the host cell's DNA (**STEP 2**); the phage DNA is protected from breakdown because it contains a modified form of cytosine that is not recognized by the phage enzyme. The entire lytic cycle, from the phage's first contact with the cell surface to cell lysis, takes only 20–30 minutes at 37°C.

Animation

Phage Lytic Cycle

image

image

The Lysogenic Cycle

Instead of lysing their host cells, many phages coexist with them in a state called lysogeny. In contrast to the lytic cycle, which kills the host cell, the **lysogenic cycle** allows replication of the phage genome without destroying the host. Phages capable of using both modes of replicating within a bacterium are called **temperate phages**. A temperate phage called lambda, written with the Greek letter λ, has been widely used in biological research. Phage λ resembles T4, but its tail has only one short tail fiber.

Infection of an *E. coli* cell by phage λ begins when the phage binds to the surface of the cell and injects its linear DNA genome **(Figure 19.6)**. Within the host, the λ DNA molecule forms a circle. What happens next depends on the replicative mode: lytic cycle or lysogenic cycle. During a lytic cycle, the viral genes immediately turn the host cell into a λ-producing factory, and the cell soon lyses and releases its virus progeny. During a lysogenic cycle, however, the λ DNA molecule is incorporated into a specific site on the *E. coli* chromosome by viral proteins that break both circular DNA molecules and join them to each other. When integrated into the bacterial chromosome in this way, the viral DNA is known as a **prophage**. One prophage gene codes for a protein that prevents transcription of most of the other prophage genes. Thus, the phage genome is mostly silent within the bacterium. Every time the *E. coli* cell prepares to divide, it replicates the phage DNA along with its own chromosome such that each daughter cell inherits a prophage. A single infected cell can quickly give rise to a large population of bacteria carrying the virus in prophage form. This mechanism enables viruses to propagate without killing the host cells on which they depend.

Figure 19.6

The lytic and lysogenic cycles of phage λ, a temperate phage.

Lytic cycle and lysogenic cycles are detailed after the lambda D N A enters the bacterial cell.

After entering the bacterial cell the lambda D N A induce a lytic cycle or a lysogenic cycle.

Lytic cycle: New phage D N A and proteins are synthesized and self-assemble into phages. The cell lyses, releasing phages. The phage attaches to a host cell and injects its D N A through the tail fiber. The Phage D N A circularizes.

Lysogenic cycle: Phage D N A integrates into the bacterial chromosome, becoming a prophage. The bacterium reproduces normally, copying the prophage and transmitting it to daughter cells. Many cell divisions produce a large population of bacteria infected with the prophage. Occasionally, a daughter cell with prophage exits the bacterial chromosome, initiating a lytic cycle.

After entering the bacterial cell and circularizing, the λ DNA can immediately initiate the production of a large number of progeny phages (lytic cycle) or integrate into the bacterial chromosome (lysogenic cycle). In most cases, phage λ follows the lytic pathway, which is similar to that detailed in Figure 19.5. However, once a lysogenic cycle begins, the prophage may be carried in the host cell's chromosome for many generations. Phage λ has one main tail fiber, which is short.

Animation

Phage Lysogenic and Lytic Cycles

image

image

The term *lysogenic* signifies that prophages are capable of generating active phages that lyse their host cells. This occurs when the λ genome (or that of another temperate phage) is induced to exit the bacterial chromosome and initiate a lytic cycle. An environmental signal, such as a certain chemical or high-energy radiation, usually triggers the switchover from the lysogenic to the lytic mode.

In addition to the gene for the viral protein that prevents transcription, a few other prophage genes may be expressed during lysogeny. Expression of these genes may alter the host's phenotype, a phenomenon that can have important medical significance. For example, the three species of bacteria that cause the human diseases diphtheria, botulism, and scarlet fever would not be so harmful to humans without certain prophage genes that cause the host bacteria to make toxins. And the difference between the *E. coli* strain in our intestines and the O157:H7 strain that has caused several deaths by food poisoning appears to be the presence of toxin genes of prophages in the O157:H7 strain.

Bacterial Defenses Against Phages

After reading about the lytic cycle, you may have wondered why phages haven't exterminated all bacteria. Lysogeny is one major reason why bacteria have been spared from extinction caused by phages. Bacteria also have their own defenses against phages. First, natural selection favors bacterial mutants with surface proteins that are no longer recognized as receptors by a particular type of phage. Second, when phage DNA does enter a bacterium, the DNA often is identified as foreign and cut up by cellular enzymes called **restriction enzymes**, which are so named because they *restrict* a phage's ability to replicate within the bacterium. (Restriction enzymes are used in molecular biology and DNA cloning techniques; see Concept 20.1.) The bacterium's own DNA is methylated in a way that prevents attack by its own restriction enzymes. A third defense is a system present in both bacteria and archaea called the *CRISPR-Cas system*, which you learned about in Concept 17.5.

HHMI BioInteractive Video

Jennifer Doudna: CRISPR-Cas9 was First Discovered in Bacteria

image

image

HHMI BioInteractive Video

Jennifer Doudna: Science is a Serendipitous Endeavor

image

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The CRISPR-Cas system was discovered during a study of repetitive DNA sequences present in the genomes of many prokaryotes. These sequences, which puzzled scientists, were named clustered regularly interspaced short palindromic repeats (CRISPRs) because each sequence reads the same forward and backward (a palindrome), with different stretches of "spacer DNA" in between the repeats. At first, scientists assumed the spacer DNA sequences were random and meaningless, but analysis by several research groups showed that each spacer sequence corresponded to DNA from a particular phage that had infected the cell. Further studies revealed that particular nuclease proteins interact with the CRISPR region. These nucleases, called Cas (CRISPR-associated) proteins, can identify and cut phage DNA, thereby defending the bacterium against phage infection.

When a phage infects a bacterial cell that has the CRISPR-Cas system, the DNA of the invading phage is stored, integrated into the genome between two repeat sequences **(Figure 19.7)**. If the cell survives the infection, any further attempt by the same type of phage to infect this cell (or its offspring) triggers transcription of the CRISPR region into RNA molecules. These RNAs are cut into pieces and then bound by Cas proteins, such as the Cas9 protein (see Figure 17.28). The Cas protein uses a portion of the phage-related RNA as a homing device to identify the invading phage DNA and cut it, leading to its destruction.

Figure 19.7

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The CRISPR–Cas system: a type of bacterial immune system.
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Drawing summarizes the C R I S P R-C a s system in five steps.

The steps are as follows.

1.Infection by a phage triggers transcription of the C R I S P R region of the bacterial D N A. This region consists of D N A from phages that previously infected the cell, separated by repeats.

2. The R N A transcript is processed into short R N A strands. Here we focus on R N A that is complementary to the D N A of the invading phage.
3. Each short R N A strand binds to a Cas protein, forming a complex. (Multiple complexes are formed.)
4. Complementary R N A binds to D N A from the invading phage. The Cas protein then cuts the phage D N A.
5. After being cut, the entire phage D N A molecule is degraded and can no longer be replicated.

Interview

****Interview with Francisco Mojica: Discovering CRISPR in bacteria

A photograph of Francisco Mojica, a Spanish molecular biologist.

Just as natural selection favors bacteria that have receptors altered by mutation or that have enzymes that cut phage DNA, it also favors phage mutants that can bind to altered receptors or that are resistant to enzymes. Thus, the bacterium-phage relationship is in constant evolutionary flux.

Replicative Cycles of Animal Viruses

Everyone has suffered from viral infections, whether cold sores, influenza, or the common cold. Like all viruses, those that cause illness in humans and other animals can replicate only inside host cells. Many variations on the basic scheme of viral infection and replication are represented among the animal viruses. Key variables are the nature of the viral genome (double- or single-stranded DNA or RNA) and the presence or absence of an envelope.

Whereas there are only a few bacteriophages that have an envelope or RNA genome, many animal viruses have both. In fact, nearly all animal viruses with RNA genomes have an envelope, as do some with DNA genomes. Rather than consider all the mechanisms of viral infection and replication, we will focus first on the roles of viral envelopes and then on the functioning of RNA as the genetic material of many animal viruses.

Viral Envelopes

An animal virus equipped with an envelope—that is, a membranous outer layer—uses it to enter the host cell. Protruding from the outer surface of this envelope are viral glycoproteins that bind to specific receptor molecules on the surface of a host cell. For example, the Spike protein on SARS-CoV-2 binds to a protein

called ACE2 on the surface of human airway (and other) cells. **Figure 19.8** outlines the events in the replicative cycle of an enveloped virus with an RNA genome. Ribosomes bound to the endoplasmic reticulum (ER) of the host cell make the protein parts of the envelope glycoproteins; cellular enzymes in the ER and Golgi apparatus then add the sugars. The resulting viral glycoproteins, embedded in membrane derived from the host cell, are transported to the cell surface. In a process much like exocytosis (see Figure 7.20), new viral capsids are wrapped in membrane as they bud from the cell. In other words, the viral envelope is usually derived from the host cell's plasma membrane, although all or most of the molecules of this membrane are specified by viral genes. The enveloped viruses are now free to infect other cells. This replicative cycle does not necessarily kill the host cell, in contrast to the lytic cycles of phages.

Figure 19.8

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The replicative cycle of an enveloped RNA virus.
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Replicative cycle of an enveloped R N A virus showing each of the stages and activities during the cycle.

Shown here is a virus with a single-stranded RNA genome that functions as a template for synthesis of mRNA.

Replicative cycle of an enveloped R N A virus: 1) Glycoproteins on the viral envelop bind to specific receptor molecules on the host cell, promoting viral uptake by the cell. 2) The capsid and viral genome enter the cell. Digestion of the capsid by cellular enzymes releases the viral genome (R N A). 3) The viral genome functions as a template for synthesis of complementary R N A strands by a viral R N A polymerase. 4) New copies of viral genome R N A are made using the complementary R N A strands as templates. 5) Complementary R N A strands also function as m R N A, which is translated into both capsid proteins (in the cytosol) and glycoproteins for the viral envelope (in the ER and Golgi apparatus). 6) Vesicles transport envelope glycoproteins to the plasma membrane. 7) A capsid assembles around each viral genome molecule. 8) Each new virus buds from the cell, its envelope studded with viral glycoproteins embedded in membrane derived from the host cell.

Some viruses have envelopes that are not derived from plasma membrane. Herpesviruses, for example, are temporarily cloaked in membrane derived from the nuclear envelope of the host; they then shed this membrane in the cytoplasm and acquire a new envelope made from membrane of the Golgi apparatus. These viruses have a double-stranded DNA genome and replicate within the host cell nucleus, using a combination of viral and cellular enzymes to replicate and transcribe their DNA. In the case of herpesviruses, copies of the viral DNA can remain behind as mini-chromosomes in the nuclei of certain nerve cells. There they remain latent until some sort of physical or emotional stress triggers a new round of active virus production. The infection of other cells by these new viruses causes the blisters characteristic of herpes, such as cold sores or

genital sores. Once someone acquires a herpesvirus infection, flare-ups may recur throughout the person's life.

Viral Genetic Material

Table 19.1 shows the common classification system for animal viruses, which is based on their genetic material: double- or single-stranded DNA, or double- or single-stranded RNA, with further classification of RNA viruses. Although some phages and most plant viruses are RNA viruses, the broadest variety of RNA genomes is found among the viruses that infect animals. There are three types of single-stranded RNA genomes found in animal viruses (classes IV–VI in Table 19.1). The genome of class IV viruses can directly serve as mRNA and thus can be translated into viral protein immediately after infection. (SARS-CoV-2 has this type of genome.) Figure 19.8 shows a virus of class V, in which the RNA genome serves instead as a *template* for mRNA synthesis. (This type is found in influenza viruses.) The RNA genome is transcribed into complementary RNA strands, which function both as mRNA and as templates for the synthesis of additional copies of genomic RNA. All viruses that use an RNA genome as a template for mRNA transcription require RNA → DNA synthesis. These viruses use a viral enzyme capable of carrying out this process; there are no such enzymes in most cells. The enzyme used in this process is encoded by the viral genome. After the protein is synthesized, it is packaged during viral self-assembly with the genome inside the viral capsid.

Table 19.1

Classes of Animal Viruses

A table titled Table 19.1 Classes of animal viruses illustrates class/ family, presence or absence of envelope, examples of viruses causing human diseases.

Class/Family	Envelope?	Examples that Cause Human Diseases
Roman numeral 1. Double-stranded D N A (d s D N A)		
Adenovirus (see figure 19.3b)	No	Respiratory viruses
Papillomavirus	No	Warts, cervical cancer
Polyomavirus	No	Tumors
Herpesvirus	Yes	Herpes simplex Roman numeral 1 and 2 (cold sores, genital sores); varicella zoster (shingles, chicken pox); Epstein-Barr virus (mononucleosis) Burkitt's lymphoma)

Poxvirus	Yes	Smallpox virus; cowpox virus
Roman numeral 2. Single-stranded D N A (s s D N A)		
Parvovirus	No	B 19 parvovirus (mild rash)
Roman numeral 3. Double-stranded R N A (d s R N A)		
Reovirus	No	Rotavirus (diarrhea); Colorado tick fever virus
Roman numeral 4. Single-stranded D N A (s s D N A; serves as m R N A		
Picornavirus	No	Rhinovirus (common cold); poliovirus; hepatitis A virus; other intestinal viruses
Coronavirus	Yes	Severe acute respiratory syndrome (S A R S); Middle east respiratory syndrome (M E R S)
Flavivirus	Yes	Zika virus (see figure 19.10c); yellow fever virus; dengue virus; West Nile virus; hepatitis C virus
Togavirus	Yes	Chikungunya virus (see figure 19.10b); rubella virus; equine encephalitis viruses
Roman numeral 5. s s R N A; serves as template for m R N A synthesis		
Filovirus	Yes	Ebola virus (haemorrhagic fever; see figure 19.10a)
Orthomyxovirus	Yes	Influenza virus (see figure 19.3c)
Paramyxovirus	Yes	Measles virus; mumps virus
Rhabdovirus	Yes	Rabies virus

Roman numeral 6. s s R N A; serves as template for D N A synthesis		
Retrovirus	Yes	Human immunodeficiency virus (H I V/ A I D S; see figure 19.9); human T- lymphotropic virus type-1 (H T L V-1) (leukemia)

The RNA animal viruses with the most complicated replicative cycles are the **retroviruses** (class VI). These viruses have an enzyme called **reverse transcriptase** that transcribes an RNA template into a DNA copy, an RNA → DNA information flow that is the opposite of the usual direction. This unusual phenomenon is the source of the name retroviruses (*retro* means "backward"). Of particular medical importance is **HIV(human immunodeficiency virus)**, the retrovirus shown in Figure 19.1 that causes **AIDS(acquired immunodeficiency syndrome)**. HIV and other retroviruses are enveloped viruses that contain two identical molecules of single-stranded RNA and two molecules of reverse transcriptase.

The HIV replicative cycle (illustrated in **Figure 19.9**) is typical of a retrovirus. After HIV enters a host cell, its reverse transcriptase molecules are released into the cytoplasm, where they catalyze synthesis of viral DNA. The newly made viral DNA then enters the cell's nucleus and integrates into the DNA of a chromosome. The integrated viral DNA, called a **provirus**, never leaves the host's genome, remaining a permanent resident of the cell. (Recall that a prophage, in contrast, leaves the host's genome at the start of a lytic cycle.) The RNA polymerase of the host transcribes the proviral DNA into RNA molecules, which can function both as mRNA for the synthesis of viral proteins and as genomes for the new viruses that will be assembled and released from the cell. In Concept 43.4, we describe how HIV causes the deterioration of the immune system that occurs in AIDS.

Figure 19.9

The replicative cycle of HIV, the retrovirus that causes AIDS.

Detailed overview of the replicative cycle of H I V in 10 steps.

Replicative cycle of H I V.

1. The H I V envelope glycoproteins enable the virus to bind to specific receptors on certain white blood cells. Glycoproteins are shown on the outside of the viral envelope and the capsid is inside with two identical strands of R N A and reverse transcriptase.

2. The virus fuses with the cell's plasma membrane. The capsid proteins are removed, releasing the viral proteins and R N A.

3. Reverse transcriptase catalyzes the synthesis of a D N A strand complementary to the viral R N A.
4. Reverse transcriptase catalyzes the synthesis of a second D N A strand complementary to the first.
5. The double-stranded D N A is incorporated as a provirus into the cell's D N A.
6. Proviral genes are transcribed into R N A molecules, which serve as genomes for progeny viruses and as m R N As for translation into viral protein.
7. The viral proteins include capsid proteins and reverse transcriptase (made in the cytosol) and envelope glycoproteins (made in the ER).
8. Vesicles transport the glycoproteins to the cell's plasma membrane.
9. Capsids are assembled around viral genomes and reverse transcriptase molecules.
10. New viruses, with viral envelope glycoproteins, bud from the host cell. Side pictures show H I V entering a cell and leaving a cell (scale 0.25 micrometers).

The photos (artificially colored TEMs) show HIV entering and leaving a human white blood cell. See Figure 7.8 for the cell-surface proteins that act as receptors for HIV. Note in **STEP 6** that DNA synthesized from the viral RNA genome is integrated as a provirus into the host cell chromosomal DNA, a characteristic unique to retroviruses.

MAKE CONNECTIONS *Describe what is known about binding of HIV to immune system cells. (See*Figure 7.8*.)* How was this discovered?*

*For suggested answer, see*Appendix A*.*

HHMI Biointeractive Animation

HIV Replicative Cycle

image

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Animation

Retrovirus (HIV) Replicative Cycle

Evolution of Viruses

EVOLUTION We began this chapter by asking whether or not viruses are alive. Viruses do not really fit our definition of living organisms. An isolated virus is biologically inert, unable to replicate its genes or regenerate its own ATP. Yet it has a genetic program written in the universal language of life. Do we think of viruses as nature's most complex associations of molecules or as the simplest forms of life? Either way, we must bend our usual definitions. Although viruses cannot replicate or carry out metabolic activities independently, their use of the genetic code makes it hard to deny their evolutionary connection to the living world.

How did viruses originate? Viruses have been found that infect every form of life—not only bacteria, animals, and plants, but also archaea, fungi, and algae and other protists. Researchers estimate that each milliliter (one-fifth of a teaspoon) of seawater contains between one and 100 million viruses, ten times the number of microorganisms! Because viruses depend on cells for their own propagation, it seems likely that viruses are not the descendants of precellular forms of life but evolved—possibly multiple times—*after* the first cells appeared. Most molecular biologists favor the hypothesis that viruses originated from naked bits of cellular nucleic acids that moved from one cell to another, perhaps via injured cell surfaces. The evolution of genes coding for capsid proteins may have allowed viruses to bind cell membranes, thus facilitating the infection of uninjured cells.

Candidates for the original sources of viral genomes include plasmids and transposons. *Plasmids* are small, circular DNA molecules found in bacteria and in the unicellular fungi called yeasts. Plasmids exist apart from the genome, can replicate independently of the genome, and are occasionally transferred between cells. *Transposons* are DNA segments that can move from one location to another within a cell's genome. Thus, plasmids, transposons, and viruses all share an important feature: They are *mobile genetic elements*. (We'll discuss plasmids in more detail in Concepts 20.1 and 27.2 and transposons in Concept 21.4.)

Consistent with this notion of pieces of DNA shuttling from cell to cell is the observation that a viral genome can have more in common with the genome of its host than with the genomes of viruses that infect other hosts. Indeed, some viral genes are essentially identical to genes of the host.

Debate about the origin of viruses was reinvigorated about 20 years ago when an extremely large virus was discovered: Mimivirus is a double-stranded DNA (dsDNA) virus with an icosahedral capsid that is 400 nm in diameter, the size of a small bacterium. Its genome contains 1.2 million bases (Mb)—about 100 times as many as the influenza virus genome—and an estimated 1,000 genes. Perhaps the most surprising aspect of mimivirus, however, was that its genome included genes previously found only in cellular genomes. Some of these genes code for proteins involved in translation, DNA repair, protein folding, and polysaccharide synthesis. Whether mimivirus evolved *before* the first cells and then developed an exploitative relationship with them or evolved more recently and simply scavenged genes from its hosts is not yet settled.

In the past decade, several even larger viruses have been discovered that cannot be classified with any existing known virus. One such virus is 1μm(1,000 nm) in diameter, with a dsDNA genome of around 2–

2.5 Mb, larger than that of some small eukaryotes. What's more, over 90% of its 2,000 or so genes are unrelated to cellular genes, inspiring the name it was given, pandoravirus. The number of genes in pandoraviruses varies from 1,500 to 2,500 genes. A second virus, called *Pithovirus sibericum*, with a diameter of 1.5μm and 500 genes, was discovered in permanently frozen soil in Siberia. This virus, once thawed, was able to infect an amoeba after being frozen for 30,000 years! How these and all other viruses fit in the tree of life is an intriguing, unresolved question.

The ongoing evolutionary relationship between viruses and the genomes of their host cells is an association that continues to make viruses very useful experimental systems in molecular biology. Knowledge about viruses also allows many practical applications, since viruses have a tremendous impact on all organisms through their ability to cause disease.

Concept Check 19.2

- Compare the effect on the host cell of a lytic (virulent) phage and a lysogenic (temperate) phage.
- **MAKE CONNECTIONS** Compare the CRISPR-Cas system with the miRNA system discussed in Concept 18.3, including their mechanisms and their functions.
- **MAKE CONNECTIONS** The RNA virus in Figure 19.8 has a viral RNA polymerase that functions in step 3 of the virus's replicative cycle. Compare this with a cellular RNA polymerase in terms of template and overall function (see Figure 17.10).
- Why is HIV called a retrovirus?
- **VISUAL SKILLS** Looking at Figure 19.9, imagine you are a researcher trying to combat HIV infection. What molecular processes could you attempt to block?

*For suggested answers, see*Appendix A.**

Concept 19.3: Viruses and prions are formidable pathogens in animals and plants

Diseases caused by viral infections afflict humans, agricultural crops, and livestock worldwide. Other smaller, less complex entities known as prions also cause disease in animals. We'll first consider animal viruses.

Viral Diseases in Animals

A viral infection can produce symptoms by a number of different mechanisms. Viruses may damage or kill cells by causing the release of hydrolytic enzymes from lysosomes. Some viruses cause infected cells to produce toxins that lead to disease symptoms, and some have molecular components that are toxic, such as envelope proteins. How much damage a virus causes depends partly on the ability of the infected tissue to regenerate by cell division. People usually recover completely from colds because the epithelium of the respiratory tract, which the viruses infect, can efficiently repair itself. In contrast, damage inflicted by poliovirus to mature nerve cells is permanent because these cells do not divide and usually cannot be replaced. Many of the temporary symptoms associated with viral infections, such as fever and body aches, actually result from the body's own efforts to defend itself against infection rather than from cell death caused by the virus.

The immune system is a critical part of the body's natural defenses (see Chapter 43). It is also the basis for the major medical tool used to prevent viral infections—vaccines. A **vaccine** is a harmless component of a pathogen that stimulates the immune system to mount defenses against the harmful pathogen. Smallpox, a viral disease that was once a devastating scourge in many parts of the world, was eradicated by 1980 due to a vaccination program carried out by the World Health Organization (WHO). The very narrow host range of the smallpox virus—it infects only humans—was a critical factor in the success of this program. Similar worldwide vaccination campaigns are under way to eradicate polio, the incidence of which has dropped by 99%, and measles. Although an effective vaccine exists for measles, a large outbreak occurred in the Pacific Northwest in 2019, correlated with lower vaccination rates in that region. Effective vaccines are also available to protect against rubella, mumps, hepatitis B, and a number of other viral diseases. Less than a year after SARS-CoV-2 was identified as the cause of COVID-19, multiple vaccines were developed and approved for use. Many nations immediately initiated programs to administer shots rapidly to their populations. Unfortunately, many less-resourced countries lagged behind in receiving needed vaccine doses, one factor contributing to the persistence of the pandemic.

Although vaccines can prevent some viral illnesses, medical care can do little, at present, to cure most viral infections once they occur. The antibiotics that help us recover from bacterial infections are powerless against viruses. Antibiotics kill bacteria by inhibiting enzymes specific to bacteria but have little or no effect on eukaryotic or virally encoded enzymes. However, the few enzymes that are encoded only by viruses have provided targets for other drugs. Most antiviral drugs resemble nucleosides and thus interfere with viral nucleic acid synthesis. One such drug is acyclovir, which impedes herpesvirus replication by inhibiting the viral polymerase that synthesizes viral DNA but not the eukaryotic one. Similarly, azidothymidine (AZT) curbs HIV replication by interfering with the synthesis of DNA by reverse transcriptase. In the past 30 years, much effort has gone into developing drugs to treat HIV. Currently, multidrug treatments, called *antiretroviral therapy* or *ART*, are considered to be most effective. Such treatments commonly include a combination of two nucleoside mimics and a protease inhibitor, which interferes with an enzyme required for assembly of the viruses. Multidrug treatments originally involved taking up to 20 pills multiple times per day but now usually consist of a single daily tablet. Another effective treatment involves a drug called

maraviroc, which blocks a protein on the surface of human immune cells that helps bind the HIV virus (see Figure 7.8). This drug has also been used successfully to prevent infection in individuals who either have been exposed to, or are at risk of exposure to, HIV. In a similar approach, the current vaccines against SARS-CoV-2 are designed to raise an immune response against the Spike protein on the viral surface, blocking it from binding to the cell surface receptor during infection.

BBC Video

Know Your Enemy: Bacteria vs. Viruses

image

image

HHMI BioInteractive Video

Treating HIV Infection with a Reverse Transcriptase Inhibitor

image

Play

image

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Emerging Viral Diseases

Viruses that suddenly become apparent are often referred to as *emerging viruses*. HIV, the AIDS virus, is a classic example: The earliest HIV infections of humans occurred in sub-Saharan Africa, perhaps as early as the 1920s or 1930s, with outbreaks in the 1960s and 1970s. In the United States, the first cases were reported in the early 1980s among young gay men in California and New York. In 1982, the Centers for Disease Control named the disease AIDS. Another example of an emerging virus is West Nile virus, which causes encephalitis, inflammation of the brain. This appeared in North America in 1999 and has spread to 49 U.S. states, by 2019 resulting in over 50,000 cases and about 2,000 deaths.

The deadly Ebola virus **(Figure 19.10a)**, recognized initially in 1976 in central Africa, is one of several emerging viruses that cause *hemorrhagic fever*, an often fatal illness characterized by fever, vomiting, massive bleeding, and circulatory system collapse. In 2014, a widespread outbreak, or **epidemic**, of Ebola virus occurred. By 2016, it had resulted in over 11,000 deaths. In 2017, 2018, and 2019, smaller outbreaks occurred in the Democratic Republic of the Congo.

Figure 19.10

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Emerging viruses.
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Three photos of emerging viruses.

The viruses are as follows.

Part a. Ebola viruses budding from a monkey cell (colorized S E M). Scale 500 nanometers.

Part b. Chikungunya viruses emerging from a cell in the upper left and packing together (colorized T E M). Scale 200 nanometers.

Part c. Computer-generated image of a Zika virus, based on a technique called cryo-electron microscopy. Scale 20 nanometers.

The mosquito-borne chikungunya virus **(Figure 19.10b)** causes an acute illness with fever, rashes, and persistent joint pain. Chikungunya has long been considered a tropical virus, but it has now appeared in Italy, France, and Spain. The Zika virus **(Figure 19.10c)** was first observed in Uganda in 1947, but for decades only a few cases occurred per year. In the spring of 2015, however, it became an emerging virus when it caused a large outbreak in Brazil. Although symptoms of Zika are often mild, the infection of pregnant individuals was correlated with a striking increase in the number of babies born with abnormally small brains, a condition called microcephaly. Zika is a mosquito-borne flavivirus (like West Nile virus) that infects neural cells, posing a particular danger to fetal brain development.

HHMI BioInteractive Video

Interview with Katie Walter

image

image

HHMI BioInteractive Video

Genetically Modified Mosquitoes

image

image

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Where do these new strains of virus come from? One cause of rapidly emerging viral diseases in humans is mutation of existing viruses into new viruses that can spread more easily. RNA viruses have a high rate of mutation because viral RNA polymerases do not proofread and correct errors in replicating their RNA genomes. Some mutations change existing viruses into new viral strains that can cause disease, even in people immune to the original virus. A well-known related example is how three or four mutations causing changes in a surface protein of a cat virus (feline panleukopenia virus) resulted in the emergence in 1978 of canine parvovirus, a very contagious deadly virus infecting dogs.

A second cause of the emergence of viral diseases is the spread of a viral disease from a small, isolated human population. AIDS went unnamed and virtually unnoticed for decades before spreading around the world, accelerated by affordable international travel. HIV is transmitted through exchange of bodily fluids; this happens, for instance, through blood transfusions, unprotected sexual intercourse, and reuse of needles to inject intravenous drugs. In African countries, lack of resources and stigmatization of infected individuals initially hastened the spreading of AIDS. In the United States, AIDS was seen at first as only affecting gay men. Societal prejudice toward this community led to a lack of resources to investigate and stem the spread of the disease, which contributed to its worldwide spread.

Interview

Interview with David Satcher: The role of the CDC in recognizing AIDS and in public health

David Satcher.

A third cause of new viral diseases in humans is the spread of existing viruses from other animals. Scientists estimate that about three-quarters of new human diseases originate in this way. Animals infected with a virus that can be transmitted to humans are said to be a natural reservoir for that virus. HIV is an example, as scientists believe that it originated from a version of the virus found in chimpanzees in central Africa, after people ate chimpanzee meat for food and were infected by exposure to chimpanzee blood. In a more recent example, many scientists think that bats passed SARS-CoV-2 to humans, possibly via an intermediate host.

In general, flu epidemics provide an instructive example of these three causes of emerging viruses. There are three types of influenza virus: types B and C, which infect only humans and have never caused an epidemic, and type A, which infects a wide range of animals, including birds, pigs, horses, and humans. The influenza type A viruses present in pigs and wild and domestic birds are potential emerging viruses that represent a long-term threat to human health.

A case in point is the H5N1 strain of avian influenza virus, which is highly contagious and deadly in birds. The first transmission from birds to humans was documented in Hong Kong in 1997. Since then, about 700 people have been infected, with an alarming mortality rate of greater than 50%. The high mortality rate is

partly because H5N1 is very different from strains of influenza that have circulated among humans for a long time. Individuals are therefore not able to mount a strong immune response against viruses like H5N1.

Different strains of influenza A are given standardized names; for example, the name H5N1 identifies which forms of two viral surface proteins are present—hemagglutinin and neuraminidase (HA and NA, respectively; see the glycoprotein spikes in Figure 19.3c). These two proteins together help determine the host range and severity of disease caused by each virus. As of 2017, 18 types of hemagglutinin, a protein that helps the flu virus attach to host cells, and 11 types of neuraminidase, an enzyme that helps release new virus particles from infected cells, have been identified. All possible combinations of HA and NA have been found in some waterbirds. In 2018, researchers using advanced microscopy techniques reported that the HA protein is quite dynamic, stretching toward the target host cell, retracting, then refolding to a new shape and re-approaching the target cell.

Although deadly, the H5N1 strain has not yet caused an epidemic because nearly all cases have been the result of transmission from birds to people, rather than person-to-person. Epidemics occur when genetic changes allow a new viral strain to be easily transmitted between humans. An event like this occurred in 2009, when a strain of influenza virus (H1N1) appeared that was very different from the virus that causes the seasonal flu. The H1N1 influenza virus spread rapidly, prompting WHO to declare a global epidemic, or **pandemic**. Within half a year, the disease had reached 207 countries, infecting over 600,000 people and killing almost 8,000. In a more recent example, the first case of COVID-19 was reported in Wuhan, China, in late December 2019, and on March 11, 2020, WHO declared it a pandemic. Within the next 18 months, virtually every country was battling SARS-CoV-2.

In addition to the 2009 H1N1 pandemic, influenza A strains have caused three other major flu epidemics among humans in the last 100 years. The most notable of these was the “Spanish flu” pandemic of 1918–1919, which killed 40–50 million people worldwide. In the **Scientific Skills Exercise**, you’ll analyze genetic changes in variants of the 2009 H1N1 influenza virus and correlate them with spread of the disease.

Scientific Skills Exercise

Analyzing a Sequence-Based Phylogenetic Tree to Understand Viral Evolution

AP Science Practices 2, 4, 5, 6

How Can Sequence Data Be Used to Track Flu Virus Evolution? In 2009, an influenza A H1N1 virus caused a pandemic, and the virus has continued to resurface in outbreaks across the world. Researchers in Taiwan were curious about why the virus kept appearing despite widespread flu vaccine

initiatives. They hypothesized that newly evolved variant strains of the H1N1 virus were able to evade human immune system defenses. To test this hypothesis, they needed to determine if each wave of flu infection was caused by a different H1N1 variant strain.

A nurse injects a patient's arm with a vaccine.

*H1N1**flu vaccination***

How the Experiment Was Done Scientists obtained the genome sequences for 4,703 virus isolates collected from patients with H1N1 flu in Taiwan, each named by type/location/identifying number/year. They compared the sequences in different strains for the viral hemagglutinin (HA) gene, and based on mutations that had occurred, arranged the isolates into a phylogenetic tree (see Figure 26.5 for information on how to read phylogenetic trees).

Data from the Experiment In the phylogenetic tree, each branch tip is one variant strain of the H1N1 virus with a unique HA gene sequence. The tree is a way to visualize a working hypothesis about the evolutionary relationships between H1N1 variants.

A phylogenetic tree.

Parts are as follows. Phylogenetic tree: each branch tip is one variant strain of the H 1 N 1 virus with a unique H A gene sequence. From the left, there are three main branch points (nodes) splitting. Starting at the top of the tree as drawn and moving down through the tree, passing through five nodes from the left and sharing the last node is the following.

A/California/07/2009 (Group 1) and

- A/Taiwan/1164/2010 (Group 3)

Passing through four nodes is:

- A/Taiwan/T1773/2009 (Group 6)

From the third node are (all Group 7, top to bottom as drawn):

- A/Taiwan/T1338/2009
- A/Taiwan/T0724/2009
- A/Taiwan/T1821/2009
- A/Taiwan/937/2009
- A/Taiwan/T1339/2009

Then another node to:

- A/Taiwan/940/2009 and

- A/Taiwan/7418/2009

Then back to the third node for:

- A/Taiwan/8575/2009

Followed by another node branching from the third node to:

- 2 nodes from 3rd node: A/Taiwan/4909/2009 and A/Taiwan/8542/2009

And two nodes branching from third node to (both Group 9):

- A/Taiwan/1018/2011 and
- A/Taiwan/552/2001

Then back to third node for:

- A/Taiwan/2826/2009 (Group 7)

Second node from left branches to:

- A/Taiwan/T0826/2009 (Group 7)

And branching from the first node from the left:

- Through 2 additional nodes: A/Taiwan/1017/2009 (Group 8) and A/Taiwan/7873/2009 (Group 8)

Then, back to branching from the first node are:

- A/Taiwan/11706/2009 (Group 8)
- A/Taiwan/6078/2009 (Group 8)
- A/Taiwan/6341/2009 (Group 8)
- A/Taiwan/6200/2009 (Group 8)

Then an additional node branches to:

- A/Taiwan/5270/2010 (Group 8-1) and
- A/Taiwan/3994/2010 (Group 8-1)

Below, a long branch from the left goes to one node branching to

- A/Taiwan/2649/2011 and A/Taiwan/1102/2011 (both Group 10)

Below which another region of heavy branching starts at the left and goes, through 3 nodes to (all Group 11):

- A/Taiwan/4501/2011 and
- A/Taiwan/67/2011

Then through two nodes from the left is:

- A/Taiwan/1749/2011

Below, from a different branch point but also through 2 nodes from the left is:

- A/Taiwan/4611/2011

Below, through three nodes is:

- A/Taiwan/5506/2011

Through four nodes is:

- A/Taiwan/1150/2011

Through 5 nodes is:

- A/Taiwan/2883/2011

And through a 6th node (sharing this node) are:

- A/Taiwan/842/2010 and
- A/Taiwan/3697/2011

INTERPRET THE DATA

- The more closely connected two variants are in the tree, the more alike they are in terms of HA gene sequence. Each fork in a branch, called a node, shows where two lineages separate due to different accumulated mutations. The length of the branches is a measure of how many sequence differences there are between the variants, indicating how distantly related they are. Referring to the phylogenetic tree, which variants are more closely related to each other: A/Taiwan/1018/2011 and A/Taiwan/552/2011 or A/Taiwan/1018/2011 and A/Taiwan/8542/2009? Explain your answer.
- The scientists arranged the branches into groups made up of one ancestral variant and all of its descendant, mutated variants. They are color-coded in the figure. Using group 11 as an example, trace the lineage of its variants.
-
    - Do all of the nodes have the same number of branches or branch tips?
- Are all of the branches in the group the same length?
- What do these results indicate?
- The graph shows the number of isolates collected (each from an ill patient) on the y-axis and the month and year that the isolates were collected on the x-axis. Each group of variants is plotted separately with a line color that matches the tree diagram.
-
    - Which group of variants was the earliest to cause the first wave of H1N1 flu in over 100

patients in Taiwan?

- After a group of variants had a peak number of infections, did members of that same group cause another (later) wave of infection?
- One variant in group 1 (green, uppermost branch) was used to make a vaccine that was distributed very early in the pandemic. Based on the graphed data, does it look like the vaccine was effective?

A graph of viral variants.

Scientists graphed the number of isolates by the month and year of isolate collection to show the period in which each viral variant was actively causing illness in people.

Data from J.-R. Yang et al., New variants and age shift to high fatality groups contribute to severe successive waves in the 2009 influenza pandemic in Taiwan, *PLoS ONE*6(11): e28288 (2011).

Graph of the number of viral isolates versus months between 2009 and 2011. Y-axis label is number of viral isolates from 0 to 800. 4 waves of the viral isolates are shown.

Wave 1: May to October, 2009 which includes Groups 1, 3, 6 with a peak of about 20 and Group 7 with a peak of about 700 – both peaks occur in August;

Wave 2: October to February 2010 which includes Group 8 with a peak of about 625 in November;

Interwave: February 2010 to November 2010 which includes Group 8-1 with a peak of about 50 in August (all other groups are flat);

Wave 3: December 2010 to April 2011 includes Group 9 (peak about 75), Group 10 (peak about 250), and Group 11 (peak about 500) all of which peak in January.

- Groups 9, 10, and 11 all had H1N1 variants that caused a large number of infections at the same time in Taiwan. Does this mean that the scientists' hypothesis, that new variants cause new waves of infection, was incorrect? Explain your answer.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

The disease caused by H1N1 was originally called "swine flu" because parts of the viral genome were very similar to strains of influenza in pigs. However, studies revealed that the virus was not transmitted from pigs to humans. Instead, the story was more complex: H1N1 was a unique combination of swine, avian, and human influenza genes that allowed it to spread among humans.

Influenza viruses can change quickly because they have a genome made up of eight segments of RNA rather than a single RNA molecule (see Figure 19.3c). When an animal like a pig or a bird is infected with multiple strains of influenza virus, the RNA molecules making up the viral genomes can mix and match (*reassort*)

during viral assembly, resulting in new genetic combinations. If a flu virus from pigs recombines with viruses that circulate widely among humans, it may acquire the ability to spread easily from person to person, dramatically increasing the potential for a major human outbreak. Pigs are believed to have been the main hosts for recombination that led to the 2009 H1N1 flu virus.

Influenza viruses also have a high rate of mutation, for reasons mentioned earlier. Coupled with reassortments, these mutations can lead to the emergence of a viral strain from animals that can infect human cells. Scientists are worried that the H5N1 strain will evolve in a way that enables it to spread as easily as the H1N1 strain. This would represent another major global health threat like that of the 1918 influenza pandemic.

How easily could this capability evolve in the H5N1 strain? In 2011, scientists working with ferrets, small mammals used as model organisms for the human flu, found that only a few mutations of the avian flu virus were sufficient to allow infection of cells in the human nasal cavity and windpipe. Furthermore, when the scientists transferred nasal swabs serially from ferret to ferret, the virus became transmissible through the air. Reports of this startling discovery ignited a firestorm of debate about whether to publish the results. Ultimately, the scientific community decided the benefits of preventing pandemics outweighed the risks of the information being used for harmful purposes, and the work was published in 2012.

In fact, we have observed a similar phenomenon in SARS-CoV-2 during the COVID-19 pandemic. Widespread SARS-CoV-2 infection of many individuals has allowed random mutations to occur, and evolutionary processes have led to new, more easily transmissible strains of the virus emerging and becoming predominant. Clearly, the COVID-19 pandemic has sharply highlighted the crucial importance of understanding the evolution and functioning of viruses.

Normal seasonal flu viruses (including influenza types A and B) are not considered emerging viruses because variations of seasonal flu viruses have been circulating among humans for long enough that most components are recognized by the immune system. However, these viruses still undergo mutation and reassortment of genome segments, and variations of the HA protein are used each year to generate vaccines against the strains predicted most likely to occur the following year.

As we have seen, emerging viruses are usually existing viruses that mutate, spread more widely in the current host species, or spread to new host species. Changes in host behavior or environmental changes can increase the viral traffic responsible for emerging diseases. For instance, new roads built through remote areas can allow viruses to spread between previously isolated human populations. Also, the destruction of forests to expand cropland can bring humans into contact with animals that host infectious viruses. Finally, genetic mutations and changes in host ranges can allow viruses to jump between species. Many viruses are transmitted by mosquitoes. A dramatic expansion of the disease caused by the chikungunya virus occurred in the mid-2000s when a mutation allowed it to infect not only the mosquito species *Aedes aegypti* but also

the related *Aedes albopictus*. Insecticides and mosquito netting over beds are crucial tools in public health attempts to prevent diseases carried by mosquitoes **(Figure 19.11)**.

Figure 19.11

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Netting as protection against virus-carrying mosquitoes.
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Photo of baby sleeping inside mosquito netting. Close-up photo of a mosquito biting human skin.

Recently, scientists have become concerned about the possible effects of climate change on worldwide viral transmission. Dengue fever, also mosquito-borne, has appeared in Florida and Portugal, regions where it had not been seen before. The possibility that global climate change has allowed mosquito species carrying these viruses to expand their ranges and interact more is troubling because of the increased chance of a mutation allowing a virus to jump to a new host. This is an area of active research by scientists applying climate change models to what is known about the habitat requirements of mosquito species.

Prions: Proteins as Infectious Agents

The viruses discussed in this chapter are infectious agents that spread diseases, and their genetic material is composed of nucleic acids, whose ability to be replicated is well known. Surprisingly, there are also *proteins* that are infectious. Proteins called **prions** appear to cause degenerative brain diseases in various animal species. These diseases include scrapie in sheep; mad cow disease, which plagued the European beef industry about 20 years ago; and Creutzfeldt-Jakob disease in humans, which has caused the death of some 175 people in the United Kingdom since 1996. Prions can be transmitted in food, as may occur when people eat beef from cattle with mad cow disease. Kuru, another human disease caused by prions, was identified in the early 1900s among the South Fore indigenous people of New Guinea. When a kuru epidemic peaked there in the 1960s, scientists at first thought the disease had a genetic basis because family members also often contracted the disease. Eventually, however, investigations revealed a different story: After a death, family members practiced ritual cannibalism, eating organs of the deceased, and prions were transmitted primarily in brain tissue. Women got kuru more often than men because men ate the more “prestigious” organs, like the heart, while women and children ate the brains.

Two characteristics of prions are especially alarming. First, prions act very slowly, with an incubation period of at least ten years before symptoms develop. The lengthy incubation period prevents sources of infection from being identified until long after the first cases appear, allowing many more infections to occur. Second, prions are not destroyed or deactivated by heating to normal cooking temperatures. To date, there is no known cure for prion diseases, and the only hope for developing effective treatments lies in understanding the process of infection.

How can a protein, which cannot replicate itself, be a transmissible pathogen? According to the leading model, a prion is a misfolded form of a protein normally present in brain cells. When the prion gets into a cell containing the normal form of the protein, the prion somehow converts normal protein molecules to the misfolded prion versions. Several prions then aggregate into a complex that can convert other normal proteins to prions, which join the chain **(Figure 19.13)**. Prion aggregation interferes with normal cellular functions and causes disease symptoms. This model was greeted with much skepticism when it was first proposed by Stanley Prusiner in the early 1980s, but it is now widely accepted. Prusiner was awarded the Nobel Prize in 1997 for his work on prions. He has also proposed that prions are involved in neurodegenerative diseases such as Alzheimer's and Parkinson's disease. There are many outstanding questions about these small infectious agents.

Figure 19.13

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Model for how prions propagate.
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Prion contacts a normally folded version of the same protein which then assumes the abnormal shape of the prion. This can continue until large aggregates of prions are formed.

Prions are misfolded versions of normal brain proteins. When a prion contacts a normally folded version of the same protein, it may induce the normal protein to assume the abnormal shape. The resulting chain reaction may continue until high levels of prion aggregation cause cellular malfunction and eventual degeneration of the brain.

Animation

Prions: Characteristics

image

Play

image

Animation

Prions: Diseases

image

Play

image

Concept Check 19.3

- Describe two ways in which a preexisting virus can become an emerging virus.
- Contrast horizontal and vertical transmission of viruses in plants.
- **WHAT IF?** TMV has been isolated from virtually all commercial tobacco products. Why, then, is TMV infection not an additional hazard for smokers?

*For suggested answers, see*Appendix A.**

Viral Diseases in Plants

More than 2,000 types of viral diseases of plants are known, accounting for an annual loss of over $30 billion worldwide due to destruction of crops. Common signs of viral infection include bleached or brown spots on leaves and fruits **(Figure 19.12)**, stunted growth, and damaged flowers or roots, all of which can diminish the yield and quality of crops.

Figure 19.12

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Immature tomato infected by a virus.
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Photo of an underdeveloped pale tomato with blotchy brown spots.

Plant viruses have the same basic structure and mode of replication as animal viruses. Most known plant viruses, including tobacco mosaic virus (TMV), have an RNA genome. Many have a helical capsid, like TMV, while others have an icosahedral capsid (see Figure 19.3b).

Viral diseases of plants spread by two major routes. In the first route, *horizontal transmission*, an external source infects the plant. Because the invading virus must get past the plant's outer protective layer of cells (the epidermis), a plant becomes more susceptible to viral infections if it has been damaged by wind, injury, or herbivores. Herbivores, especially insects, pose a double threat because they can also carry viruses, transmitting disease from plant to plant. Moreover, gardeners may transmit plant viruses inadvertently on pruning shears and other tools. The other route of viral infection is *vertical transmission*, in which a plant inherits a viral infection from a parent. Vertical transmission can occur in asexual propagation (for example, through cuttings) or in sexual reproduction via infected seeds.

Once a virus enters a plant cell and begins replicating, viral genomes and associated proteins can spread throughout the plant through plasmodesmata, the cytoplasmic connections that penetrate the walls between adjacent plant cells (see Figure 36.19). The passage of viral macromolecules from cell to cell is facilitated by virally encoded proteins that cause enlargement of plasmodesmata. Scientists have not yet devised cures for

most viral plant diseases, so research efforts are focused largely on reducing disease transmission and on breeding resistant varieties of crop plants.